

Reference excerpt 02-001

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arXiv:1604.05177v1 [physics.med-ph] 18 Apr 2016 notes series Quantitative analysis of jugular venous pulse obtained by using a general-purpose ultrasound scanner Francesco Sisini Department of Physics and Earth Sciences, University of Ferrara, Via Saragat 1, 44122 Ferrara, Italy Published on 14 April 2016. *Ikken hissatsu* () means something like to annihilate at one blow. This document is part of a series of notes each one targeting a single goal. Each note has to annihilate at one blow ! CONDITIONS OF USE This is a self-published methodological note distributed under the Creative Commons Attribution License (<http://creativecommons.org/licenses/by/4.0/>), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited. The note contains an original reasoning of mine and the goal to share thoughts and methodologies, not results. Therefore before using the contents of these notes, everyone is invited to verify the accuracy of the assumptions and conclusions. INTRODUCTION Jugular venous pulse (JVP) examination dates back to more than a century ago [Mackenzie(1902), Mackay(1947), Applefeld(1990), Kalmanson et al. (1972)], but only recently it has been shown that the trace of the JVP can be obtained by means of an ultrasound scanner (US) equipped with a commercial linear probe.[Sisini et al.(2015), Sahani et al.(2015)] The method proposed needs a video clip of a few seconds obtained from the transverse scan of the neck, where the internal jugular vein (IJV) is clearly visible. [Sisini et al. (2015)] The cross sectional area of the IJV (CSA) is measured on